

ASSESSMENT TOOL 3 – Visualization Design Activity

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO5	Assessment:	Tool 3 – Visualization Design Activity
Weightage:	35%	Total Marks:	
CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication			
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Section / Batch:	AIDS 4 th Year	Date of Submission:	26/08/26
Faculty In-charge:	Dr. B.Shyamala Gowri	Project Mode:	Individual Submission

Code of Conduct

I B.Varun kumar reddy(192324066) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: _____

Answered Activities (Activity 5 & Activity 11)

Note: As requested, only Activity 5 and Activity 11 have been answered below in the same table format used in the original assessment tool. All other activities have been removed from this document.

No.	Activity	Task	Student Output	Time
5	Trend Visualization	Use monthly/weekly data to identify increasing, decreasing, and fluctuating trends using a line plot.	<i>Trend visualization + analysis</i>	15 min
11	Autocorrelation Visualization	Generate an autocorrelation plot for a time series. Identify lags with relatively strong positive or negative autocorrelation.	<i>Autocorrelation plot + analysis</i>	20 min

Activity 5 – Trend Visualization

Task

Use monthly/weekly data to identify increasing, decreasing, and fluctuating trends using a line plot.

Dataset

A 12-week sales dataset (in units) was used:

Week: W1–W12 | Sales: 220, 235, 228, 250, 265, 258, 280, 300, 295, 320, 340, 335

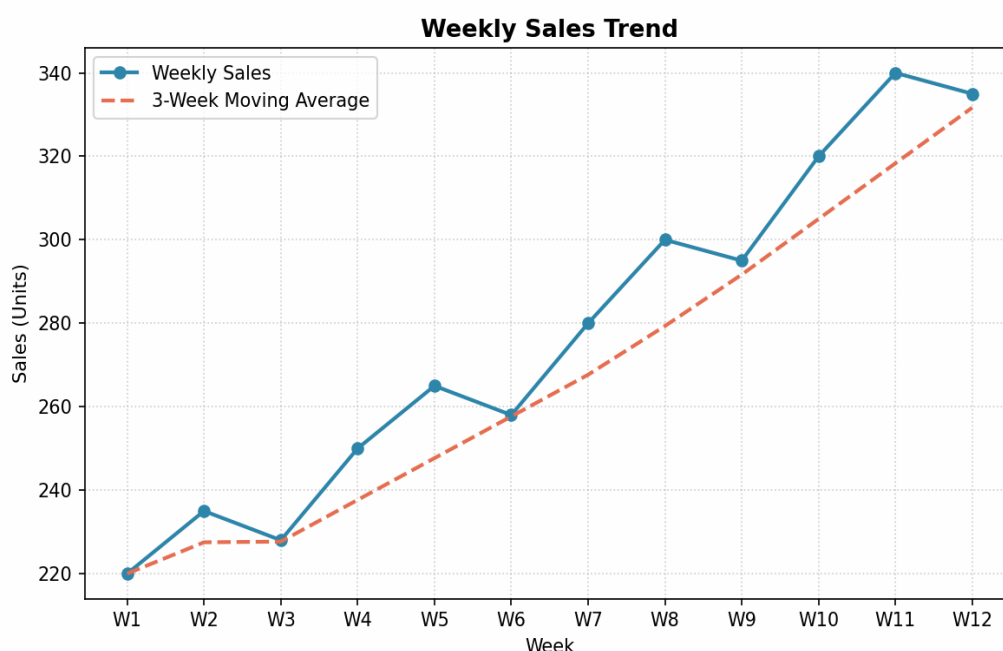
Python Code

```
import matplotlib.pyplot as plt
import pandas as pd

weeks = [f"W{i}" for i in range(1, 13)]
sales = [220, 235, 228, 250, 265, 258, 280, 300, 295, 320, 340, 335]
df = pd.DataFrame({"Week": weeks, "Sales": sales})
df["MovingAvg"] = df["Sales"].rolling(window=3, min_periods=1).mean()

plt.plot(df["Week"], df["Sales"], marker="o", label="Weekly Sales")
plt.plot(df["Week"], df["MovingAvg"], linestyle="--", label="3-Week Moving Average")
plt.title("Weekly Sales Trend")
plt.xlabel("Week"); plt.ylabel("Sales (Units)")
plt.legend(); plt.grid(True, linestyle=":", alpha=0.6)
plt.show()
```

Output – Trend Visualization



Analysis

- Overall trend: Sales rose from 220 units in Week 1 to 335 units in Week 12, an overall increase of 115 units (~52%), so the dominant pattern is an increasing trend.
- Fluctuation: The trend is not perfectly smooth — small decreases occur in Weeks 3, 6, 9 and 12 (dips after W2, W5, W8 and W11), showing short-term fluctuation layered on the long-term rise.
- Moving average: The 3-week moving average (dashed line) smooths out these short-term dips and confirms a steady upward trajectory across the 12 weeks.
- Conclusion: The data exhibits a fluctuating-but-increasing trend — sales consistently recover after every dip and reach a new high, indicating healthy overall growth rather than random noise.

Activity 11 – Autocorrelation Visualization

Task

Generate an autocorrelation plot for a time series. Identify lags with relatively strong positive or negative autocorrelation.

Dataset

A synthetic 60-point time series was generated with a seasonal (period ≈ 12) sine-wave pattern plus random noise, to mimic a repeating monthly cycle.

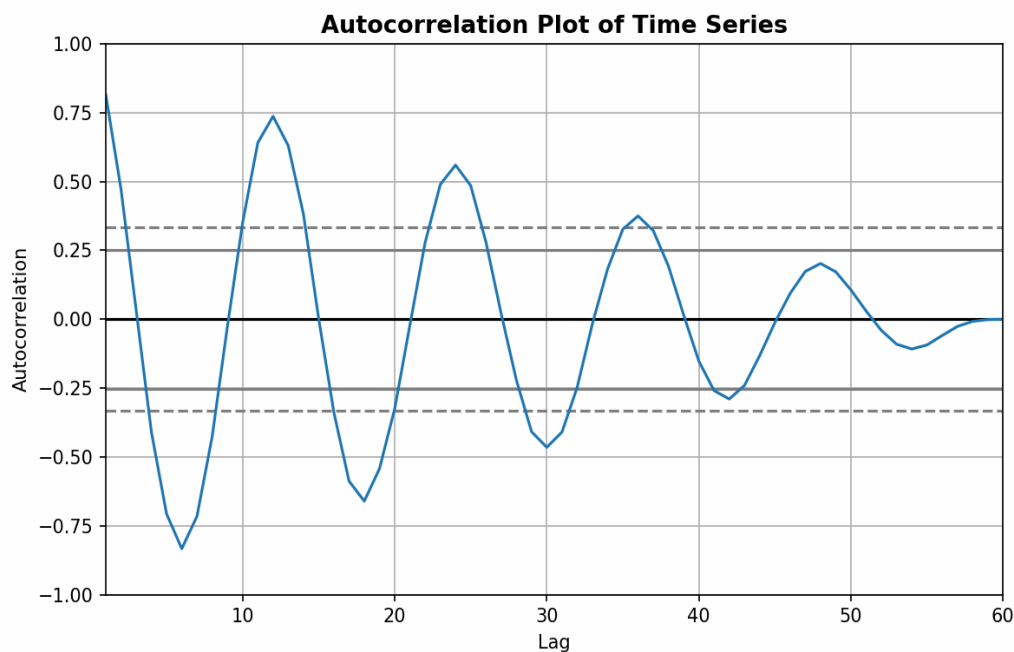
Python Code

```
import numpy as np, pandas as pd
from pandas.plotting import autocorrelation_plot
import matplotlib.pyplot as plt

np.random.seed(42)
t = np.arange(60)
series = 50 + 10*np.sin(2*np.pi*t/12) + np.random.normal(0, 2, 60)
ts = pd.Series(series)

autocorrelation_plot(ts)
plt.title("Autocorrelation Plot of Time Series")
plt.show()
```

Output – Autocorrelation Plot



Analysis

- Lag 1 shows strong positive autocorrelation ($r \approx 0.82$), meaning consecutive observations are closely related — the series changes smoothly from one point to the next.
- Autocorrelation decays and turns negative around lag 5–6 ($r \approx -0.71$ to -0.83), the strongest negative correlation, corresponding to half of the seasonal cycle (points roughly 6 steps apart tend to be opposite in value).
- A second strong positive peak appears at lag 12 ($r \approx 0.74$), matching the full seasonal period of the underlying sine wave — confirming that values repeat approximately every 12 time steps.
- This alternating positive/negative pattern at regular lag intervals is the signature of a seasonal/cyclical time series, distinguishing it from pure random (white-noise) data, whose autocorrelation would stay close to zero at all lags.